

SECTION 1: Identification of the substance/mixture and of the company/undertaking**1.1 Product identifier**

Product name Alphasyn EP 460
Product code 462251-BE02
SDS # 462251
Product type Liquid.

1.2 Relevant identified uses of the substance or mixture and uses advised against

**Use of the substance/
mixture** Lubricant.
For specific application advice see appropriate Technical Data Sheet or consult our company representative.

1.3 Details of the supplier of the safety data sheet

Supplier Lubricants UK Limited,
Chertsey Road,
Sunbury On Thames,
Middlesex,
TW16 7BP

+44 (0)345 600 8125
E-mail address MSDSadvise@bp.com

1.4 Emergency telephone number

**EMERGENCY
TELEPHONE NUMBER** Carechem: +44 (0) 1235 239 670 (24/7)

SECTION 2: Hazards identification**2.1 Classification of the substance or mixture**

Product definition Mixture
Classification according to Regulation (EC) No. 1272/2008 [CLP/GHS]
☒ Not classified.

See sections 11 and 12 for more detailed information on health effects and symptoms and environmental hazards.

2.2 Label elements

Signal word No signal word.
Hazard statements ☒ No known significant effects or critical hazards.
Precautionary statements
Prevention ☒ Not applicable.
Response Not applicable.
Storage Not applicable.
Disposal ☒ Not applicable.
Hazardous ingredients Not applicable.
**Supplemental label
elements** ☒ Contains Polysulfides, di-tert-Bu, Polysulfides, di-tert-dodecyl and Amines, C10-14-tert-alkyl.
May produce an allergic reaction.
Safety data sheet available on request.

EU Regulation (EC) No. 1907/2006 (REACH)

**Annex XVII - Restrictions
on the manufacture,
placing on the market
and use of certain
dangerous substances,
mixtures and articles** Not applicable.

Product name Alphasyn EP 460	Product code 462251-BE02	Page: 1/14
Version 2	Date of issue 16 September 2025	Format United Kingdom (UK) (United Kingdom)
Date of previous issue 13 August 2024.		Language ENGLISH

SECTION 2: Hazards identification

Special packaging requirements

Containers to be fitted with child-resistant fastenings	Not applicable.
Tactile warning of danger	Not applicable.

2.3 Other hazards

Product meets the criteria for PBT or vPvB according to Regulation (EC) No. 1907/2006, Annex XIII	This mixture does not contain any substances that are assessed to be a PBT or a vPvB.
Product meets the criteria for endocrine disrupting properties according to Regulation (EC) No. 1907/2006.	☒ This substance/mixture does not contain any components that are considered to have endocrine disrupting properties.
Other hazards which do not result in classification	Defatting to the skin. Experimental data on one or more of the components has been used to determine all or part of the hazard classification of this product.

SECTION 3: Composition/information on ingredients

3.2 Mixtures

Product definition	Mixture
Synthetic base stock. Proprietary performance additives.	

Product/ingredient name	Identifiers	%	Classification	Specific Conc. Limits, M-factors and ATEs	Type
☒ Polysulfides, di-tert-Bu	REACH #: 01-2119540515-43 EC: 273-103-3 CAS: 68937-96-2	≤3	Skin Sens. 1B, H317 Aquatic Acute 1, H400 Aquatic Chronic 1, H410	Skin Sens. 1, H317: C ≥ 6% M [Acute] = 1 M [Chronic] = 1	[1]
Polysulphides, di-tert-dodecyl	REACH #: 01-2119540516-41 EC: 270-335-7 CAS: 68425-15-0	<1	Skin Sens. 1B, H317	-	[1]
Amines, C10-14-tert-alkyl	REACH #: 01-2119456798-18 EC: 701-175-2 CAS: -	<0.1	Acute Tox. 4, H302 Acute Tox. 3, H311 Acute Tox. 2, H330 Skin Corr. 1B, H314 Eye Dam. 1, H318 Skin Sens. 1A, H317 Aquatic Acute 1, H400 Aquatic Chronic 1, H410	ATE [Oral] = 500 mg/kg ATE [Dermal] = 300 mg/kg ATE [Inhalation (vapours)] = 0.5 mg/l M [Acute] = 1 M [Chronic] = 1	[1]

See Section 16 for the full text of the H statements declared above.

[1] Substance classified with a health or environmental hazard

SECTION 4: First aid measures

4.1 Description of first aid measures

Eye contact	In case of contact, immediately flush eyes with plenty of water for at least 15 minutes. Eyelids should be held away from the eyeball to ensure thorough rinsing. Check for and remove any contact lenses. Get medical attention.
Skin contact	Wash skin thoroughly with soap and water or use recognised skin cleanser. Remove contaminated clothing and shoes. Wash clothing before reuse. Clean shoes thoroughly before reuse. Get medical attention if irritation develops.
Inhalation	If inhaled, remove to fresh air. Get medical attention if symptoms occur.
Ingestion	☒ Do not induce vomiting unless directed to do so by medical personnel. Get medical attention if symptoms occur.
Protection of first-aiders	☒ No action shall be taken involving any personal risk or without suitable training.

4.2 Most important symptoms and effects, both acute and delayed

See Section 11 for more detailed information on health effects and symptoms.

Potential acute health effects

Product name	Alphasyn EP 460	Product code	462251-BE02	Page:	2/14
Version	2	Date of issue	16 September 2025	Format	United Kingdom (UK) (United Kingdom)
Date of previous issue	13 August 2024.			Language	ENGLISH

SECTION 4: First aid measures

Inhalation	Vapour inhalation under ambient conditions is not normally a problem due to low vapour pressure.
Ingestion	No known significant effects or critical hazards.
Skin contact	Defatting to the skin. May cause skin dryness and irritation.
Eye contact	No known significant effects or critical hazards.

Delayed and immediate effects as well as chronic effects from short and long-term exposure

Inhalation	Overexposure to the inhalation of airborne droplets or aerosols may cause irritation of the respiratory tract.
Ingestion	Ingestion of large quantities may cause nausea and diarrhoea.
Skin contact	Prolonged or repeated contact can defat the skin and lead to irritation and/or dermatitis.
Eye contact	Potential risk of transient stinging or redness if accidental eye contact occurs.

4.3 Indication of any immediate medical attention and special treatment needed

Notes to physician	Treatment should in general be symptomatic and directed to relieving any effects.
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SECTION 5: Firefighting measures

5.1 Extinguishing media

Suitable extinguishing media	In case of fire, use foam, dry chemical or carbon dioxide extinguisher or spray.
Unsuitable extinguishing media	Do not use water jet. The use of a water jet may cause the fire to spread by splashing the burning product.

5.2 Special hazards arising from the substance or mixture

Hazards from the substance or mixture	In a fire or if heated, a pressure increase will occur and the container may burst.
Hazardous combustion products	Combustion products may include the following: carbon oxides (CO, CO ₂) (carbon monoxide, carbon dioxide)

5.3 Advice for firefighters

Special precautions for fire-fighters	No action shall be taken involving any personal risk or without suitable training. Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire.
Special protective equipment for fire-fighters	Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode. Clothing for fire-fighters (including helmets, protective boots and gloves) conforming to European standard EN 469 will provide a basic level of protection for chemical incidents.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

For non-emergency personnel	No action shall be taken involving any personal risk or without suitable training. Evacuate surrounding areas. Keep unnecessary and unprotected personnel from entering. Do not touch or walk through spilt material. Floors may be slippery; use care to avoid falling. Put on appropriate personal protective equipment.
For emergency responders	Specialised clothing is required to deal with the spillage, take note of any information in Section 8 on suitable and unsuitable materials. See also the information in "For non-emergency personnel".

6.2 Environmental precautions

Avoid dispersal of spilt material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air).

6.3 Methods and material for containment and cleaning up

Small spill	Stop leak if without risk. Move containers from spill area. Absorb with an inert material and place in an appropriate waste disposal container. Dispose of via a licensed waste disposal contractor.
Large spill	Stop leak if without risk. Move containers from spill area. Prevent entry into sewers, water courses, basements or confined areas. Contain and collect spillage with non-combustible, absorbent material e.g. sand, earth, vermiculite or diatomaceous earth and place in container for disposal according to local regulations. Dispose of via a licensed waste disposal contractor.

Product name Alphasyn EP 460

Product code 462251-BE02

Page: 3/14

Version 2 **Date of issue** 16 September 2025

Format United Kingdom (UK)

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Date of previous issue 13 August 2024.

(United Kingdom)

SECTION 6: Accidental release measures

6.4 Reference to other sections

See Section 1 for emergency contact information.
See Section 5 for firefighting measures.
See Section 8 for information on appropriate personal protective equipment.
See Section 12 for environmental precautions.
See Section 13 for additional waste treatment information.

SECTION 7: Handling and storage

7.1 Precautions for safe handling

Protective measures
Advice on general occupational hygiene

Put on appropriate personal protective equipment.
Eating, drinking and smoking should be prohibited in areas where this material is handled, stored and processed. Wash thoroughly after handling. Remove contaminated clothing and protective equipment before entering eating areas. See also Section 8 for additional information on hygiene measures.

7.2 Conditions for safe storage, including any incompatibilities

Store in accordance with local regulations. Store in a dry, cool and well-ventilated area, away from incompatible materials (see Section 10). Keep away from heat and direct sunlight. Keep container tightly closed and sealed until ready for use. Containers that have been opened must be carefully resealed and kept upright to prevent leakage. Store and use only in equipment/containers designed for use with this product. Do not store in unlabelled containers.

Not suitable

Prolonged exposure to elevated temperature

7.3 Specific end use(s)

Recommendations

See section 1.2 and Exposure scenarios in annex, if applicable.

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Occupational exposure limits

No exposure limit value known.

Whilst specific OELs for certain components may be shown in this section, other components may be present in any mist, vapour or dust produced. Therefore, the specific OELs may not be applicable to the product as a whole and are provided for guidance only.

Recommended monitoring procedures

Reference should be made to monitoring standards, such as the following: European Standard EN 689 (Workplace atmospheres - Guidance for the assessment of exposure by inhalation to chemical agents for comparison with limit values and measurement strategy) European Standard EN 14042 (Workplace atmospheres - Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents) European Standard EN 482 (Workplace atmospheres - General requirements for the performance of procedures for the measurement of chemical agents) Reference to national guidance documents for methods for the determination of hazardous substances will also be required.

DNELs/DMELs

Product/ingredient name

Amines, C10-14-tert-alkyl

Result

DNEL - Workers - Long term - Inhalation

12.5 mg/m³
Effects: Systemic

DNEL - Workers - Long term - Inhalation

12.1 mg/m³
Effects: Local

DNEL - General population - Long term - Inhalation

2.5 mg/m³
Effects: Systemic

DNEL - General population - Long term - Inhalation

1.2 mg/m³
Effects: Local

DNEL - General population - Long term - Oral

0.35 mg/kg bw/day
Effects: Systemic

8.2 Exposure controls

SECTION 8: Exposure controls/personal protection

Appropriate engineering controls	Provide exhaust ventilation or other engineering controls to keep the relevant airborne concentrations below their respective occupational exposure limits. All activities involving chemicals should be assessed for their risks to health, to ensure exposures are adequately controlled. Personal protective equipment should only be considered after other forms of control measures (e.g. engineering controls) have been suitably evaluated. Personal protective equipment should conform to appropriate standards, be suitable for use, be kept in good condition and properly maintained. Your supplier of personal protective equipment should be consulted for advice on selection and appropriate standards. For further information contact your national organisation for standards. The final choice of protective equipment will depend upon a risk assessment. It is important to ensure that all items of personal protective equipment are compatible.
Individual protection measures	
Hygiene measures	Wash hands, forearms and face thoroughly after handling chemical products, before eating, smoking and using the lavatory and at the end of the working period. Ensure that eyewash stations and safety showers are close to the workstation location.
Respiratory protection	In case of insufficient ventilation, wear suitable respiratory equipment. For protection against metal working fluids, respiratory protection that is classified as “resistant to oil” (class R) or oil proof (class P) should be selected where appropriate. Depending on the level of airborne contaminants, an air-purifying, half-mask respirator (with HEPA filter) including disposable (P- or R-series) (for oil mists less than 50mg/m3), or any powered, air-purifying respirator equipped with hood or helmet and HEPA filter (for oil mists less than 125 mg/m3). Where organic vapours are a potential hazard during metalworking operations, a combination particulate and organic vapour filter may be necessary. The correct choice of respiratory protection depends upon the chemicals being handled, the conditions of work and use, and the condition of the respiratory equipment. Safety procedures should be developed for each intended application. Respiratory protection equipment should therefore be chosen in consultation with the supplier/manufacturer and with a full assessment of the working conditions.
Eye/face protection	Safety glasses with side shields.
Skin protection	
Hand protection	General Information: Because specific work environments and material handling practices vary, safety procedures should be developed for each intended application. The correct choice of protective gloves depends upon the chemicals being handled, and the conditions of work and use. Most gloves provide protection for only a limited time before they must be discarded and replaced (even the best chemically resistant gloves will break down after repeated chemical exposures). Gloves should be chosen in consultation with the supplier / manufacturer and taking account of a full assessment of the working conditions. Recommended: Nitrile gloves. Breakthrough time: Breakthrough time data are generated by glove manufacturers under laboratory test conditions and represent how long a glove can be expected to provide effective permeation resistance. It is important when following breakthrough time recommendations that actual workplace conditions are taken into account. Always consult with your glove supplier for up-to-date technical information on breakthrough times for the recommended glove type. Our recommendations on the selection of gloves are as follows: Continuous contact: Gloves with a minimum breakthrough time of 240 minutes, or >480 minutes if suitable gloves can be obtained. If suitable gloves are not available to offer that level of protection, gloves with shorter breakthrough times may be acceptable as long as appropriate glove maintenance and replacement regimes are determined and adhered to. Short-term / splash protection: Recommended breakthrough times as above. It is recognised that for short-term, transient exposures, gloves with shorter breakthrough times may commonly be used. Therefore, appropriate maintenance and replacement regimes must be determined and rigorously followed. Glove Thickness:

SECTION 8: Exposure controls/personal protection

For general applications, we recommend gloves with a thickness typically greater than 0.35 mm.

It should be emphasised that glove thickness is not necessarily a good predictor of glove resistance to a specific chemical, as the permeation efficiency of the glove will be dependent on the exact composition of the glove material. Therefore, glove selection should also be based on consideration of the task requirements and knowledge of breakthrough times. Glove thickness may also vary depending on the glove manufacturer, the glove type and the glove model. Therefore, the manufacturers' technical data should always be taken into account to ensure selection of the most appropriate glove for the task.

Note: Depending on the activity being conducted, gloves of varying thickness may be required for specific tasks. For example:

- Thinner gloves (down to 0.1 mm or less) may be required where a high degree of manual dexterity is needed. However, these gloves are only likely to give short duration protection and would normally be just for single use applications, then disposed of.
- Thicker gloves (up to 3 mm or more) may be required where there is a mechanical (as well as a chemical) risk i.e. where there is abrasion or puncture potential.

Skin and body

Use of protective clothing is good industrial practice. Personal protective equipment for the body should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product. Cotton or polyester/cotton overalls will only provide protection against light superficial contamination that will not soak through to the skin. Overalls should be laundered on a regular basis. When the risk of skin exposure is high (e.g. when cleaning up spillages or if there is a risk of splashing) then chemical resistant aprons and/or impervious chemical suits and boots will be required.

Refer to standards:

- Respiratory protection: EN 529
- Gloves: EN 420, EN 374
- Eye protection: EN 166
- Filtering half-mask: EN 149
- Filtering half-mask with valve: EN 405
- Half-mask: EN 140 plus filter
- Full-face mask: EN 136 plus filter
- Particulate filters: EN 143
- Gas/combined filters: EN 14387

Environmental exposure controls

Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels.

SECTION 9: Physical and chemical properties

The conditions of measurement of all properties are at standard temperature and pressure unless otherwise indicated.

9.1 Information on basic physical and chemical properties

Physical state	Liquid.
Colour	Yellow. [Light]
Odour	Unfragranced
Odour threshold	Not available.
Melting point/freezing point	Not available.
Initial boiling point and boiling range	Not available.
Flammability	Not available.
Lower and upper explosion limit	Not available.
Flash point	Open cup: >200°C (>392°F) [Cleveland ASTM D 92]
Auto-ignition temperature	

SECTION 9: Physical and chemical properties

	<table><tr><th>Ingredient name</th><th>°C</th><th>°F</th><th>Method</th></tr><tr><td>Dec-1-ene, homopolymer, hydrogenated Dec-1-ene, oligomers, hydrogenated</td><td>343 to 369</td><td>649.4 to 696.2</td><td>ASTM D 2159</td></tr><tr><td>Dec-1-ene, homopolymer, hydrogenated Dec-1-ene, oligomers, hydrogenated</td><td>343 to 369</td><td>649.4 to 696.2</td><td>ASTM D 2159</td></tr></table>	Ingredient name	°C	°F	Method	Dec-1-ene, homopolymer, hydrogenated Dec-1-ene, oligomers, hydrogenated	343 to 369	649.4 to 696.2	ASTM D 2159	Dec-1-ene, homopolymer, hydrogenated Dec-1-ene, oligomers, hydrogenated	343 to 369	649.4 to 696.2	ASTM D 2159																							
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Dec-1-ene, homopolymer, hydrogenated Dec-1-ene, oligomers, hydrogenated	343 to 369	649.4 to 696.2	ASTM D 2159																																	
Decomposition temperature	Not available.																																			
pH	Not applicable.																																			
Kinematic viscosity	Kinematic: 467 mm²/s (467 cSt) at 40°C Kinematic: 53.2 mm²/s (53.2 cSt) at 100°C (ASTM D 445)																																			
Solubility																																				
	<table><tr><th>Media</th><th>Result</th></tr><tr><td>water</td><td>Not soluble</td></tr></table>	Media	Result	water	Not soluble																															
Media	Result																																			
water	Not soluble																																			
Partition coefficient n-octanol/ water (log value)	Not applicable.																																			
Vapour pressure	<table><tr><th rowspan="2">Ingredient name</th><th colspan="2">Vapour Pressure at 20°C</th><th colspan="3">Vapour pressure at 50°C</th></tr><tr><th>mm Hg</th><th>kPa</th><th>Method</th><th>mm Hg</th><th>kPa</th><th>Method</th></tr><tr><td>Dec-1-ene, homopolymer, hydrogenated Dec-1-ene, oligomers, hydrogenated</td><td><0.0041</td><td><0.00055</td><td>ASTM E 1194-87</td><td></td><td></td><td></td></tr><tr><td>Dec-1-ene, homopolymer, hydrogenated Dec-1-ene, oligomers, hydrogenated</td><td><0.0041</td><td><0.00055</td><td>ASTM E 1194-87</td><td></td><td></td><td></td></tr><tr><td>Polysulfides, di-tert-Bu</td><td>0.12</td><td>0.016</td><td>OECD 104</td><td></td><td></td><td></td></tr></table>			Ingredient name	Vapour Pressure at 20°C		Vapour pressure at 50°C			mm Hg	kPa	Method	mm Hg	kPa	Method	Dec-1-ene, homopolymer, hydrogenated Dec-1-ene, oligomers, hydrogenated	<0.0041	<0.00055	ASTM E 1194-87				Dec-1-ene, homopolymer, hydrogenated Dec-1-ene, oligomers, hydrogenated	<0.0041	<0.00055	ASTM E 1194-87				Polysulfides, di-tert-Bu	0.12	0.016	OECD 104			
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	mm Hg	kPa	Method	mm Hg	kPa	Method																														
Dec-1-ene, homopolymer, hydrogenated Dec-1-ene, oligomers, hydrogenated	<0.0041	<0.00055	ASTM E 1194-87																																	
Dec-1-ene, homopolymer, hydrogenated Dec-1-ene, oligomers, hydrogenated	<0.0041	<0.00055	ASTM E 1194-87																																	
Polysulfides, di-tert-Bu	0.12	0.016	OECD 104																																	
Density and/or Relative density	<1000 kg/m³ (<1 g/cm³) at 15°C																																			
Relative vapour density	Not available.																																			
Particle characteristics																																				
Median particle size	Not applicable.																																			
9.2 Other information																																				
Evaporation rate	Not available.																																			
Explosive properties	Not available.																																			
Oxidising properties	Not available.																																			
Pour point	-45 °C																																			

SECTION 10: Stability and reactivity

10.1 Reactivity	No specific test data available for this product. Refer to Conditions to avoid and Incompatible materials for additional information.
10.2 Chemical stability	The product is stable.
10.3 Possibility of hazardous reactions	Under normal conditions of storage and use, hazardous reactions will not occur. Under normal conditions of storage and use, hazardous polymerisation will not occur.
10.4 Conditions to avoid	Avoid all possible sources of ignition (spark or flame).
10.5 Incompatible materials	Reactive or incompatible with the following materials: oxidising materials.
10.6 Hazardous decomposition products	Under normal conditions of storage and use, hazardous decomposition products should not be produced.

SECTION 11: Toxicological information

11.1 Information on hazard classes as defined in Regulation (EC) No 1272/2008

Acute toxicity

Product/ingredient name

Amines, C10-14-tert-alkyl

Result

Rat - Oral - LD50
612 mg/kg
OECD 401

Rat - Dermal - LD50
251 mg/kg
OECD 402

Rat - Inhalation - LC50 Vapour
1.19 mg/l [4 hours]
OECD 403

Acute toxicity estimates

Product/ingredient name	Oral (mg/kg)	Dermal (mg/kg)	Inhalation (gases) (ppm)	Inhalation (vapours) (mg/l)	Inhalation (dusts and mists) (mg/l)
Amines, C10-14-tert-alkyl	500	300	N/A	0.5	N/A

Skin corrosion/irritation

Product/ingredient name

Amines, C10-14-tert-alkyl

Result

Rabbit - Skin - Visible necrosis

Serious eye damage/eye irritation

Product/ingredient name

Amines, C10-14-tert-alkyl

Result

Rabbit - Eyes - Visible necrosis

Respiratory corrosion/irritation

Not available.

Respiratory or skin sensitization

Product/ingredient name

Amines, C10-14-tert-alkyl

Result

Guinea pig - skin
OECD 406
Result: Sensitising

Germ cell mutagenicity

Product/ingredient name

Amines, C10-14-tert-alkyl

Result

In vitro - Bacteria
Bacterial Reverse Mutation Test
Result: Negative

In vitro - Mammal - species unspecified
In vitro Mammalian Cell Gene Mutation Test
Result: Negative

In vivo - Mammal - species unspecified
Mammalian Erythrocyte Micronucleus Test
Result: Negative

Carcinogenicity

Product name	Alphasyn EP 460	Product code	462251-BE02	Page:	8/14
Version	2	Date of issue	16 September 2025	Format	United Kingdom (UK) (United Kingdom)
Date of previous issue	13 August 2024.	Language	ENGLISH		

SECTION 11: Toxicological information

Not available.

Reproductive toxicity

Product/ingredient name

Amines, C10-14-tert-alkyl

Result

Rat - Oral

OECD 415

Maternal toxicity: Negative

Fertility effects: Negative

Developmental: Negative

Specific target organ toxicity (single exposure)

Not available.

Specific target organ toxicity (repeated exposure)

Not available.

Aspiration hazard

Not available.

Information on likely routes of exposure

Routes of entry anticipated: Oral, Dermal, Inhalation, Eyes.

Potential acute health effects

Inhalation

Vapour inhalation under ambient conditions is not normally a problem due to low vapour pressure.

Ingestion

No known significant effects or critical hazards.

Skin contact

Defatting to the skin. May cause skin dryness and irritation.

Eye contact

No known significant effects or critical hazards.

Symptoms related to the physical, chemical and toxicological characteristics

Inhalation

No specific data.

Ingestion

No specific data.

Skin contact

Adverse symptoms may include the following:
irritation
dryness
cracking

Eye contact

No specific data.

Delayed and immediate effects as well as chronic effects from short and long-term exposure

Inhalation

Overexposure to the inhalation of airborne droplets or aerosols may cause irritation of the respiratory tract.

Ingestion

Ingestion of large quantities may cause nausea and diarrhoea.

Skin contact

Prolonged or repeated contact can defat the skin and lead to irritation and/or dermatitis.

Eye contact

Potential risk of transient stinging or redness if accidental eye contact occurs.

Potential chronic health effects

Not available.

Conclusion/Summary [Product]

Not available.

General

No known significant effects or critical hazards.

Carcinogenicity

No known significant effects or critical hazards.

Mutagenicity

No known significant effects or critical hazards.

Developmental effects

No known significant effects or critical hazards.

Fertility effects

No known significant effects or critical hazards.

11.2 Information on other hazards

11.2.1 Endocrine disrupting properties

Conclusion/Summary [Product]

This substance/mixture does not contain any components that are considered to have endocrine disrupting properties.

11.2.2 Other information

Not available.

Product name	Alphasyn EP 460	Product code	462251-BE02	Page:	9/14
Version	2	Date of issue	16 September 2025	Format	United Kingdom (UK) (United Kingdom)
Date of previous issue	13 August 2024.	Language	ENGLISH		

SECTION 12: Ecological information

12.1 Toxicity

Product/ingredient name	Result
Amines, C10-14-tert-alkyl	Acute - ErC50 OECD 201 Algae 0.44 mg/l [72 hours] Acute - EC50 OECD 202 Daphnia 2.5 mg/l [48 hours] Acute - LC50 OECD 203 Fish 1.3 mg/l [96 hours] Acute - EC50 OECD 209 Micro-organism 63.5 mg/l [3 hours] Chronic - NOEC OECD 201 Algae 0.05 mg/l [72 hours]
Environmental hazards	Not classified as dangerous Product not classified for environmental effects. Based on data available for this or related materials.

12.2 Persistence and degradability

Not expected to be rapidly degradable.

Product/ingredient name	Result
Amines, C10-14-tert-alkyl	OECD 301D 21.8% [28 days] - Not readily

12.3 Bioaccumulative potential

This product is not expected to bioaccumulate through food chains in the environment.

Product/ingredient name	LogP _{ow}	BCF	Potential
Polysulfides, di-tert-Bu	5.6	-	High
Polysulfides, di-tert-dodecyl	>6.2	-	High
Amines, C10-14-tert-alkyl	2.9	-	Low

12.4 Mobility in soil

Soil/water partition coefficient

Not available.

Results of PMT and vPvM assessment

Product/ingredient name	PMT	P	M	T	vPvM	vP	vM
Polysulfides, di-tert-Bu	No	No	No	No	No	No	No
Polysulfides, di-tert-dodecyl	No	No	No	No	No	No	No
Amines, C10-14-tert-alkyl	No	No	No	No	No	No	No

Mobility	Spillages may penetrate the soil causing ground water contamination.
Conclusion/Summary	The product does not meet the criteria to be considered as a PMT or vPvM.

12.5 Results of PBT and vPvB assessment

Regulation (EC) No. 1907/2006 [REACH]

Product/ingredient name	PBT	P	B	T	vPvB	vP	vB
Polysulfides, di-tert-Bu	No	N/A	N/A	No	N/A	N/A	N/A
Polysulfides, di-tert-dodecyl	No	N/A	N/A	No	N/A	N/A	N/A
Amines, C10-14-tert-alkyl	No	N/A	N/A	No	N/A	N/A	N/A

Regulation (EC) No. 1272/2008 [CLP]

Product name Alphasyn EP 460	Product code 462251-BE02	Page: 10/14
Version 2	Date of issue 16 September 2025	Format United Kingdom (UK) (United Kingdom)
Date of previous issue 13 August 2024.		Language ENGLISH

SECTION 12: Ecological information

Product/ingredient name	PBT	P	B	T	vPvB	vP	vB
Polysulfides, di-tert-Bu	No	No	No	No	No	No	No
Polysulfides, di-tert-dodecyl	No	No	No	No	No	No	No
Amines, C10-14-tert-alkyl	No	No	No	No	No	No	No

Conclusion/Summary
Regulation (EC) No. 1272/2008
[CLP] The product does not meet the criteria to be considered as a PBT or vPvB.

12.6 Endocrine disrupting properties

Conclusion/Summary [Product] This substance/mixture does not contain any components that are considered to have endocrine disrupting properties.

Other ecological information Spills may form a film on water surfaces causing physical damage to organisms. Oxygen transfer could also be impaired.

12.7 Other adverse effects No known significant effects or critical hazards.

SECTION 13: Disposal considerations

13.1 Waste treatment methods

Product

Methods of disposal Where possible, arrange for product to be recycled. Dispose of via an authorised person/ licensed waste disposal contractor in accordance with local regulations.

Hazardous waste Yes.

European waste catalogue (EWC)

Waste code	Waste designation
13 02 06*	synthetic engine, gear and lubricating oils

However, deviation from the intended use and/or the presence of any potential contaminants may require an alternative waste disposal code to be assigned by the end user.

Packaging

Methods of disposal Where possible, arrange for product to be recycled. Dispose of via an authorised person/ licensed waste disposal contractor in accordance with local regulations.

Waste code	European waste catalogue (EWC)
15 01 10*	packaging containing residues of or contaminated by hazardous substances

Special precautions This material and its container must be disposed of in a safe way. Empty containers or liners may retain some product residues. Avoid dispersal of spilt material and runoff and contact with soil, waterways, drains and sewers.

References Commission 2014/955/EU
Directive 2008/98/EC

SECTION 14: Transport information

	ADR/RID	ADN	IMDG	IATA
14.1 UN number or ID number	Not regulated.	Not regulated.	Not regulated.	Not regulated.
14.2 UN proper shipping name	-	-	-	-
14.3 Transport hazard class(es)	-	-	-	-
14.4 Packing group	-	-	-	-
14.5 Environmental hazards	No.	No.	No.	No.
Additional information	-	-	-	-

SECTION 14: Transport information

14.6 Special precautions for user Not available.

14.7 Maritime transport in bulk according to IMO instruments Not available.

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture
[EU Regulation \(EC\) No. 1907/2006 \(REACH\)](#)

[Annex XIV - List of substances subject to authorisation](#)

[Annex XIV](#)

None of the components are listed.

[Substances of very high concern](#)

None of the components are listed.

[Annex XVII - Restrictions on the manufacture, placing on the market and use of certain dangerous substances, mixtures and articles](#)

☒ No listed substance

[Labelling](#) Not applicable.

[Other regulations](#)

[REACH Status](#)

The company, as identified in Section 1, sells this product in the EU in compliance with the current requirements of REACH.

[United States inventory \(TSCA 8b\)](#)

All components are active or exempted.

[Australia inventory \(AIIIC\)](#)

All components are listed or exempted.

[Canada inventory](#)

All components are listed or exempted.

[China inventory \(IECSC\)](#)

All components are listed or exempted.

[Japan inventory \(CSCL\)](#)

☒ At least one component is not listed.

[Korea inventory \(KECI\)](#)

All components are listed or exempted.

[Philippines inventory \(PICCS\)](#)

All components are listed or exempted.

[Taiwan Chemical Substances Inventory \(TCSI\)](#)

All components are listed or exempted.

[Explosive precursors](#) Not applicable.

[Ozone depleting substances \(EU 2024/590\)](#)

Not listed.

[Prior Informed Consent \(PIC\) \(649/2012/EU\)](#)

Not listed.

[Persistent Organic Pollutants](#)

Not listed.

[EU - Water framework directive - Priority substances](#)

None of the components are listed.

[Seveso Directive](#)

This product is not controlled under the Seveso Directive.

15.2 Chemical safety assessment

A Chemical Safety Assessment has been carried out for one or more of the substances within this mixture. A Chemical Safety Assessment has not been carried out for the mixture itself.

Product name Alphasyn EP 460

Product code 462251-BE02

Page: 12/14

Version 2 Date of issue 16 September 2025

Format United Kingdom (UK) (United Kingdom)

Language ENGLISH

Date of previous issue 13 August 2024.

SECTION 16: Other information

Abbreviations and acronyms	ADN = European Provisions concerning the International Carriage of Dangerous Goods by Inland Waterway ADR = The European Agreement concerning the International Carriage of Dangerous Goods by Road ATE = Acute Toxicity Estimate BCF = Bioconcentration Factor CAS = Chemical Abstracts Service CLP = Classification, Labelling and Packaging Regulation [Regulation (EC) No. 1272/2008] CSA = Chemical Safety Assessment CSR = Chemical Safety Report DMEL = Derived Minimal Effect Level DNEL = Derived No Effect Level EINECS = European Inventory of Existing Commercial chemical Substances ES = Exposure Scenario EUH statement = CLP-specific Hazard statement EWC = European Waste Catalogue GHS = Globally Harmonized System of Classification and Labelling of Chemicals IATA = International Air Transport Association IBC = Intermediate Bulk Container IMDG = International Maritime Dangerous Goods LogPow = logarithm of the octanol/water partition coefficient MARPOL = International Convention for the Prevention of Pollution From Ships, 1973 as modified by the Protocol of 1978. ("Marpol" = marine pollution) OECD = Organisation for Economic Co-operation and Development PBT = Persistent, Bioaccumulative and Toxic PNEC = Predicted No Effect Concentration REACH = Registration, Evaluation, Authorisation and Restriction of Chemicals Regulation [Regulation (EC) No. 1907/2006] RID = The Regulations concerning the International Carriage of Dangerous Goods by Rail RRN = REACH Registration Number SADT = Self-Accelerating Decomposition Temperature SVHC = Substances of Very High Concern STOT-RE = Specific Target Organ Toxicity - Repeated Exposure STOT-SE = Specific Target Organ Toxicity - Single Exposure TWA = Time weighted average UN = United Nations UVCB = Complex hydrocarbon substance VOC = Volatile Organic Compound vPvB = Very Persistent and Very Bioaccumulative Varies = may contain one or more of the following 64741-88-4 / RRN 01-2119488706-23, 64741-89-5 / RRN 01-2119487067-30, 64741-95-3 / RRN 01-2119487081-40, 64741-96-4/ RRN 01-2119483621-38, 64742-01-4 / RRN 01-2119488707-21, 64742-44-5 / RRN 01-2119985177-24, 64742-45-6, 64742-52-5 / RRN 01-2119467170-45, 64742-53-6 / RRN 01-2119480375-34, 64742-54-7 / RRN 01-2119484627-25, 64742-55-8 / RRN 01-2119487077-29, 64742-56-9 / RRN 01-2119480132-48, 64742-57-0 / RRN 01-2119489287-22, 64742-58-1, 64742-62-7 / RRN 01-2119480472-38, 64742-63-8, 64742-65-0 / RRN 01-2119471299-27, 64742-70-7 / RRN 01-2119487080-42, 72623-85-9 / RRN 01-2119555262-43, 72623-86-0 / RRN 01-2119474878-16, 72623-87-1 / RRN 01-2119474889-13
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Procedure used to derive the classification according to Regulation (EC) No. 1272/2008 [CLP/GHS]

Classification	Justification
Not classified.	

Full text of abbreviated H statements	317 H400 H410	May cause an allergic skin reaction. Very toxic to aquatic life. Very toxic to aquatic life with long lasting effects.
Full text of classifications [CLP/GHS]	Aquatic Acute 1 Aquatic Chronic 1 Skin Sens. 1B	SHORT-TERM (ACUTE) AQUATIC HAZARD - Category 1 LONG-TERM (CHRONIC) AQUATIC HAZARD - Category 1 SKIN SENSITISATION - Category 1B

History

Date of issue/ Date of revision	16/09/2025.
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Prepared by	Product Stewardship

Indicates information that has changed from previously issued version.

Notice to reader

Product name	Alphasyn EP 460	Product code	462251-BE02	Page:	13/14
Version	2	Date of issue	16 September 2025	Format	United Kingdom (UK) (United Kingdom)
Date of previous issue	13 August 2024.	Language	ENGLISH		

SECTION 16: Other information

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Product name	Alphasyn EP 460	Product code	462251-BE02	Page:	14/14
Version	2	Date of issue	16 September 2025	Format	United Kingdom (UK) (United Kingdom)
Date of previous issue	13 August 2024.			Language	ENGLISH